

Chapter 9 - Tax Profiles

- 1) The Framework NameScope shows us that we have an 'fw, bw $\rightarrow$ bf' stream and a 'w, bf $\rightarrow$ bw' stream. We can see also that there is a reduction and an AllReduce.

~~We also know that we have a $[4096 \times 8192]$ at $[8192]$ with output $[4096]$~~

We know that 'fw, bw $\rightarrow$ bf' lowered to $[1, 4096] = [4096, 8192] @ [8192, 1]$ meaning i.e. locally, $b=1$, $f=4096$, and $w=8192$.

The copy operation is that locally the output of the first fusion into the reduction.

We also see that 8 devices are participating in the AllReduce.

Putting it together, we have $\rightarrow$ first fusion followed by an AllReduce. But the way this occurs when we have $BD \cdot DF_x \rightarrow BF_x$ and $BF_x \cdot F_x D \rightarrow BD \{x\}$ $\rightarrow$ BD is tensor contraction AllReduce.

Since we have 8-way TP, $R=1$, $D=8192$, and $f=32,768$

then, the first fusion is $BD \cdot DF_x \rightarrow BF_x$

the second fusion is $BF_x \cdot F_x D \rightarrow BD \{x\}$,

followed by AllReduce $(BD \{x\})$.

Finally, the initial reduce seems to be ~~reducing~~ squashing the Batch Dimension.

(2)

- We see that the input embedding takes about 11ms
 - Each layer takes ~ 36 ms. 8 layers gives 2.88ms
 - The unembedding takes 4ms.
- Adding all of it up, we get about 350ms.

- Zooming in, we observe that we All (the prior to embedding). This seems highly wasteful.

Specifically, we can infer from FLO ops that $b=8$, $t=1024$, $n=$
 $v=32128$, $d=2048$.

The AllGate op shows us we are AllGather over the B dim,
 so we know B_x . ~~For~~ XLA AllGather the large [8, 1024,
 32128] unembedded ring instead of applying the ~~the~~ the
 embedding (unblock) to obtain B_x embedded state.

Next we poll for the attention responses. See fig.

No see the ~~graph~~ $\mathcal{G}$ (nodes are S, B, A, R and edges are $A \rightarrow R, R \rightarrow B, B \rightarrow S, S \rightarrow A$).
 Follow the path $S \rightarrow B \rightarrow A \rightarrow R$. The $\mathcal{G}$ is $[R, A, B, S] \cdot D[R, H, A, B, S]$.
 $\rightarrow [R, A, B, S]$. Since D is 0 , the $\mathcal{G}$ is $[R, A, B, S]$.
 D is 0 over X . An optimization gate to itself over X as we keep it for later work.

~~Late, but the AB , the AC Plane over BC axis. So
 $[B, T, H, V] \rightarrow [H, V, T] \rightarrow [B, T, D] \rightarrow [V, T, D]$. So it seems that
 the W_0 is checked over V axis.~~

In the FFW, we see an AllReduce following the
up-projection. This implies the extrinsic dimension D was
sharded over X (the AllReduce shows two replica groups)
↳ $[B, D_x][D_x, F] \rightarrow B, F \{D_x\}$. Then $[B, F][F, D_x]$
 $\rightarrow [B, D_x]$

During attention, we can actually see the AllReduce
of the QK^T matrix muls. This implies that D is once
more sharded over X (replica group $2, 4 \equiv 4, 2 (1, 2)$).
ie, we have $[B, D_x]$ and $[D_x, H, V], [D_x, H, Q], [D, H, H]$

We also AllReduce after W_O , so we conclude that
 $[H, V, D]$ as well. (replica groups are size 2 here). This implies
 $[B, H, T, V, D]$.

From there, I hypothesize we have a format model
parallelism where we are sharding ~~the~~ B over X ,
 D over X , H over V , V over V , and F over V .

(We also see an AllReduce over Y axis at the end of each Down proj,
intrinsically for sharding over V)

This is a non-standard combination of TP+FSPP. Normally,
 W_{up} is sharded over V , but here it is sharded over X .

▷ Data size = 8, $D = 2048 \times 4 = 8192$, $F = 16, 8192 \times 2 = 32,768$

▷ The MLP Block takes $\sim 2/3$ of the layer computation time.

▷ A roofline estimate for the Block D as follows:

▷ Our batch size $> 1024 \times 2 = 2048$ per chip, so for FSP+TP we are definitely compute bound.

▷
$$T_{\text{math_MLP}} = \frac{2 \times 2 \times 8 \times 1024 \times 8192 \times 32,768}{8 \times 1.97e14} = 5.58 \text{ms}$$

~~$$T_{\text{math_attn}} = 2 \times 2 \times 8 \times 1024 \times 8192 \times 16 \times 256$$~~

~~Since to generation, we are memory bound, so~~

~~$$t_{\text{comm_attn}} = \frac{2 \times 8192 \times 32 \times 256}{8.2e11} = 0.04 \text{ms}$$~~

~~$$8.2e11 \cdot 8$$~~

~~$$+ \frac{8 \times 1024 \times 16 \times 256 \times 2 \times 2 \times 2}{8.2e11 \cdot 8} \approx 0.04 \text{ms}$$~~

~~$$\sim 0.1 \text{ms}$$~~

We are computing attention bc it's parallel with FFN 240

$$T_{\text{mattn}} = \frac{4 \cdot 8 \cdot 1024 \cdot 8192 \cdot 32 \cdot 256}{1.97e4 \cdot 8}$$

$$\approx 1.4 \text{ms}$$

$$\frac{T_{\text{mattn}}}{T_{\text{mattn}}} = \frac{5.6}{1.4} = 4$$

~~vs the~~ The empirical assoc ratio is 2, indicating the default profile is underutilizing the hardware